

# BookTrendsRecSys

Helping readers pick their next book — fast, fair, and useful

Workshop 2

October 16, 2025

# Agenda

- The problem we solve
- What you get
- Quick demo preview
- Trust & fairness
- Where this fits today
- What is next and how to try it

# The Problem

- **Too many options:** for most readers, choosing a next book takes time.
- **Not sure what to trust:** ratings are messy and popularity can be misleading.
- **Staff time is limited:** librarians and curators need quick shortlists.

# Our Promise

- **A clean Top-10 list** per user — titles you can actually act on.
- **At a glance insights:** why these books appear (based on past likes).
- **Fair experience:** we watch for gaps across regions and genres.

# What You Get

- **Save time:** fewer clicks, faster choices.
- **Better fit:** suggestions follow what the reader enjoyed before.
- **Explainable:** simple visuals to see patterns in ratings.
- **Portable:** runs as a small app; no special hardware needed.

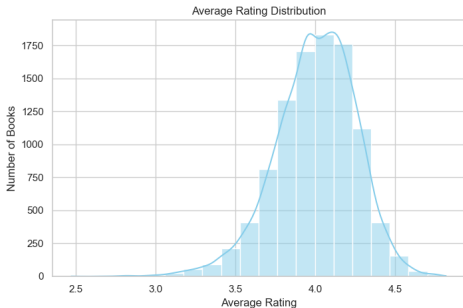
# One-Minute View of How It Works

## Behind the scenes

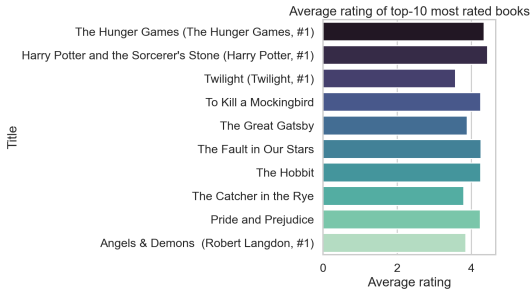
Your past ratings → we learn look-alike tastes → score many books → show the Top-10.

- Think of it as “people who liked X also liked Y” made practical.
- If the list is empty, we fall back to helpful, popular picks.

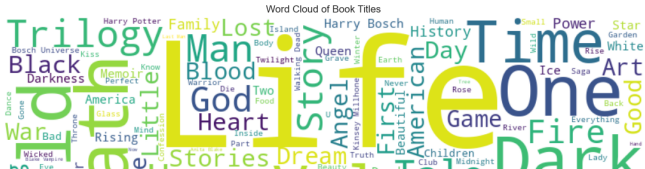
# Demo Preview



## Ratings at a glance



## Popular vs. quality overview



- **Goal:** similar experience for readers in different groups.
- **What we check:** how often the list is useful across regions or genres.
- **Today:** small gaps; we keep monitoring as data changes.



# Who Benefits

- **Readers:** quick, relevant ideas when they open the app.
- **Librarians/Curators:** ready-to-share shortlists.
- **Program managers:** basic fairness snapshot without extra tooling.

- **Front page module:** “Because you liked...” Top-10.
- **Kiosk/library desk:** fast suggestions during patron support.
- **Newsletter picks:** plug the list into monthly recommendations.

# Limitations

- Works best if the reader has **some** rating history.
- Sparse genres may need **fallback** to popularity.
- We focus on relevance first; **deep explanations** are in progress.

# What Is Next

- **Smarter cold-start:** better picks for new users.
- **Genre-aware fallback:** keep variety while staying relevant.
- **Simple reasons:** short labels like “Because you liked *Dune*”.

# How to Try It

- Open the app, enter a user ID, and (optionally) a genre.
- Review the Top-10 and the simple insights view.
- Share feedback: did this save time? did it feel relevant?

## BookTrendsRecSys

Find books you may enjoy. Friendly insights included.

[Recommendations](#) [Insights](#)

### Get book ideas

Pick a user or a book and (optionally) a genre. We'll show a short list you can skim quickly.

User index

51

- +

Genre filter

All

Show recommendations

|   | title                                        | authors                     | primary_genre | average_rating | ratings_count | score  | user_region |
|---|----------------------------------------------|-----------------------------|---------------|----------------|---------------|--------|-------------|
| 0 | Await Your Reply                             | Dan Chaon                   | ENG           | 3.54           | 10598         | 1.3466 | EMEA        |
| 1 | Good Faeries/Bad Faeries                     | Brian Froud, Terry Windling | EN-US         | 3.88           | 35068         | 1.3308 | EMEA        |
| 2 | The Killing Hour (Quincy & Rainie, #4)       | Lisa Gardner                | EN-US         | 4.16           | 17948         | 1.2887 | EMEA        |
| 3 | Out of the Silent Planet (Space Trilogy, #1) | C.S. Lewis                  | ENG           | 3.92           | 49617         | 1.2763 | EMEA        |

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# Thank You

- Questions and ideas welcome.
- Want a short pilot? We can set one up quickly.



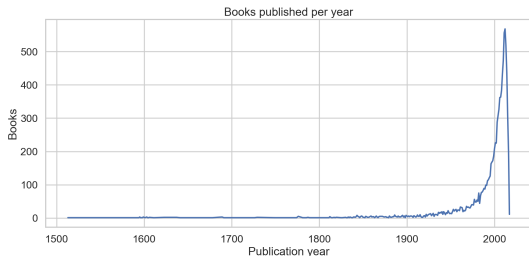
## Quick Evaluation Snapshot (Appendix)

- Precision@10: 0.0240%   Recall@10: 0.0204%   NDCG@10: 0.0249%
- Validation NDCG@10: 0.0148%
- Region gap (NDCG@10): 0.0052 pp   Genre gap: 0.0786 pp

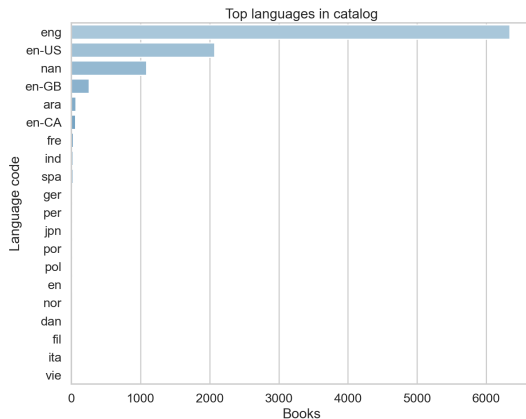
## Under the Hood (Appendix)

- Data: public ratings; train/validation split.
- Model: implicit ALS (`rank=64`, `reg=0.05`, `alpha=20`, `iters=10`).
- App: small Streamlit front end with a simple insights tab.

# Extra Visuals (Appendix)



Books per year



Top languages



## (Backup) Ranking Metrics — Formulas

For user  $u$  with ground truth  $G_u$  and top- $K$  list  $R_u = [i_1, \dots, i_K]$ :

$$\text{P@K} = \frac{1}{|U|} \sum_u \frac{|R_u \cap G_u|}{K}, \quad \text{R@K} = \frac{1}{|U|} \sum_u \frac{|R_u \cap G_u|}{|G_u|}$$

$$\text{DCG@K}(u) = \sum_{j=1}^K \frac{\mathbf{1}[i_j \in G_u]}{\log_2(j+1)}, \quad \text{NDCG@K} = \frac{1}{|U|} \sum_u \frac{\text{DCG@K}(u)}{\text{IDCG@K}(u)}$$

## (Backup) Minimal File Map

```
/scripts
|-- get_rating_of_users.py
|-- get_rating_of_books.py
|-- build_goodreads_dataset.py
|-- count_books.py
'-- goodreads_dataset.csv
/src
|-- data_prep.py  |-- als_train.py
|-- eval_ranking.py  |-- parity.py
/figs/eda, /models, /metrics, /data/processed
```